

Chapter 3: Mobile NW.

3.1 : Mobile NW : medium Access protocol,
Intranet protocol and Transport layer.

3.2 : medium Access control ;

Motivation for specialized MAC.

Introduction to multiple access technique (MAA)

3.3: mobile IP : IP packet delivery.

Agent advertisement and discovery.

Registration, Tunneling and encapsulation,
Reverse Tunneling, Routing (PDSV, DSR)

3.4 Mobile TCP.

Traditional TCP, Classical TCP improvements

like indirect TCP, Snooping TCP & mobile

TCP, Fast Retransmit / Fast recovery

Transmission / Timeout-freezing

Selective retransmission.

Mobile Network

A mobile network can be defined as a communication network that is spread out over an area around the world.

Medium Access Control

- MAC is a sublayer of data link layer (d11) in seven layer OSI network reference model.
- MAC is responsible for the transmission of data packets to and from the n/w interface card and to and from another remotely shared channel.
- MAC address is a hardware address used to uniquely identify each node of n/w.

Functions performed in the MAC Sublayer :

- ① Addressing : MAC sublayer performs the addressing of destination station and conveyance of source-station addressing information.
- ② Transparent data transfer : It performs the data transparency over data transfer of LLC, PDUs or equivalent info in Ethernet Sublayer.
- ③ Protection - MAC Sublayer Function is to protect data against error generally by means of generating and checking frame check sequence.

Access control - Control of access to the physical transmission medium from unauthorized medium access

Motivation for a specialized MAC

① Hidden and exposed terminals

* Hidden terminals

→ A sends to B, C cannot receive A.

→ C wants to send to B, C senses a "free"

medium (CS fails)

→ Collision at B, A cannot receive the

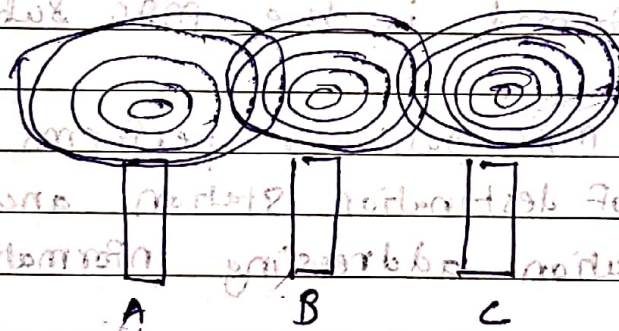
collision. (CD fails.)

→ A is "hidden" for C

or vice versa

B → A & C Exposed terminal

or vice versa



Hidden terminal - Collision

Exposed terminal - delay

② CSMA - collision detection

→ CSMA carrier sense

* Exposed terminals

- B sends to A, C wants to send to

another terminal (not A or B)

→ C has to wait, CS signals a medium

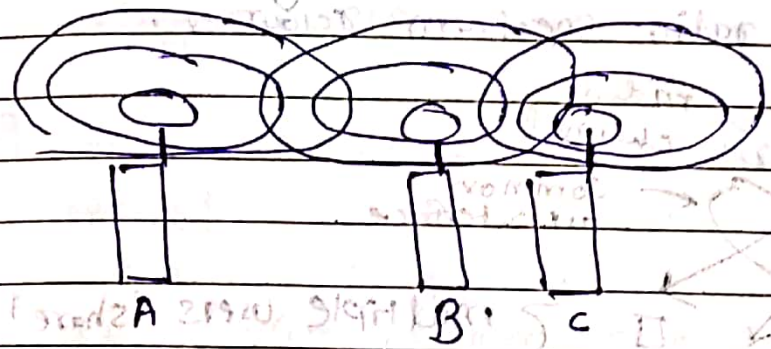
is busy

- but A is outside the radio range of C, therefore waiting is not necessary.
- C is "exposed" to B.

② Near and far terminals:

Terminal A and B send, C Receiver.

- Signal strength decreases proportional to the square of the distance.
- The signal of terminal B therefore ~~is~~ ^{is} stronger out A's signal.
- C cannot receive A.



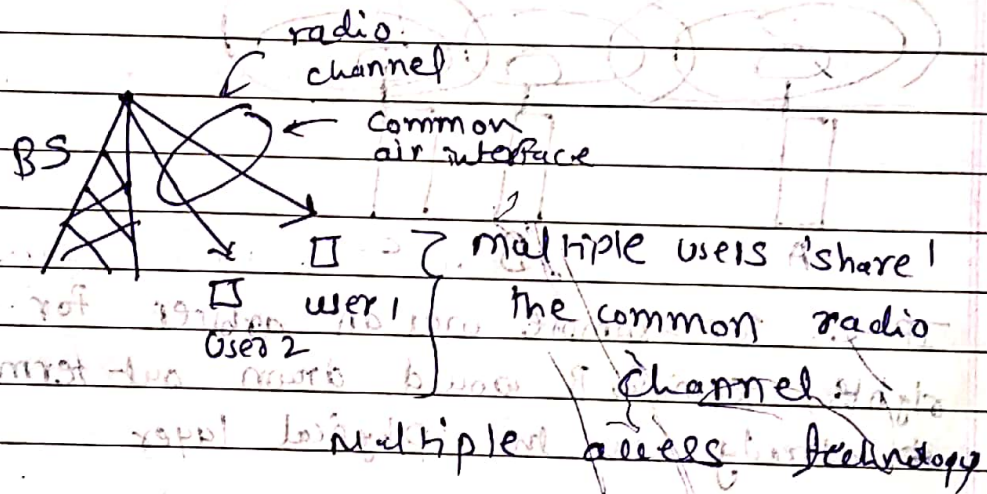
If C for example was an arbiter for sending rights, terminal B would drown out terminal A already on the physical layer.

Multiple Access Technique

- Multiple access method allows several terminals connected to the same multipoint transmission medium to transmit over it.

Example: of shared physical media are, wireless n/ws, bus n/ws, ring n/w and half duplex point-to-point links.

- Multiple access schemes to allow many users to share simultaneously a finite amount of radio spectrum resources.



Technique of multiple access

(1) Frequency division multiple access (FDMA)
frequency band is divided into channels of equal bandwidth so that each conversation is carried on a different frequency.

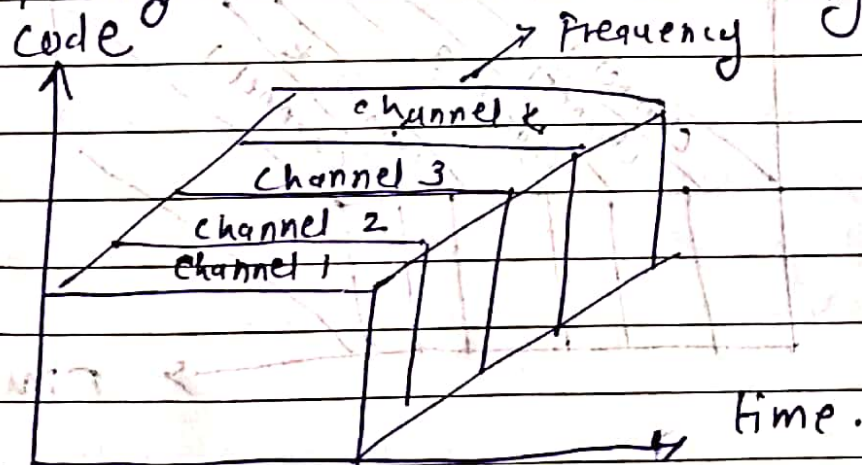
(2) A specific frequency band is given to one person and it will be received by identifying each of the frequency on the receiving end.

(3) It is often used in first generation of analog mobile phone.

(4) Each frequency station assigned a fixed frequency.

(5) In FDMA system signaling dimensions are analog. The frequency axis into non overlapping channels.

(6) Each end user assigned a different frequency channel as shown in fig.



② Time division multiple access (TDMA)

① Time Frame is divide into slots (channels).
Each user is allocated a particular time slot or channel.

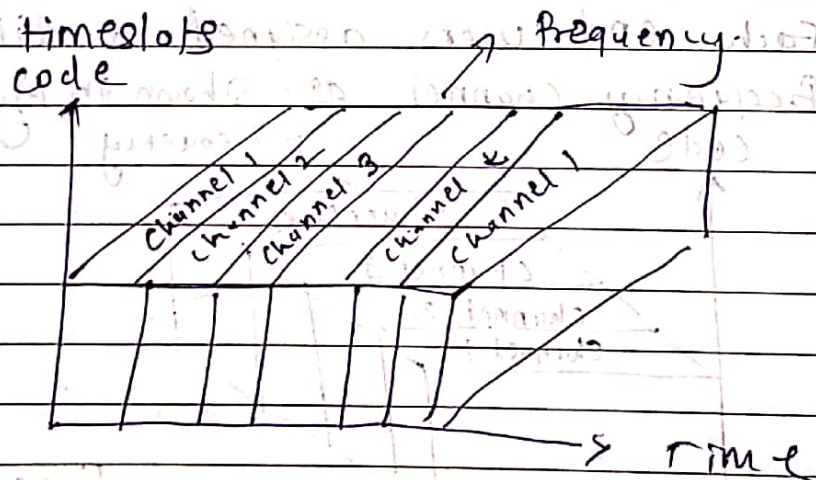
② It is digital cellular telephone communication technology.

③ It facilitates many users to share the same frequency with interference.

④ Its technology divides a signal into different timeslots and increases the data carrying capacity.

⑤ In TDMA System Signaling dimensions are divided along Time axis into non overlapping channels.

⑥ Each end users assigned a different timeslots

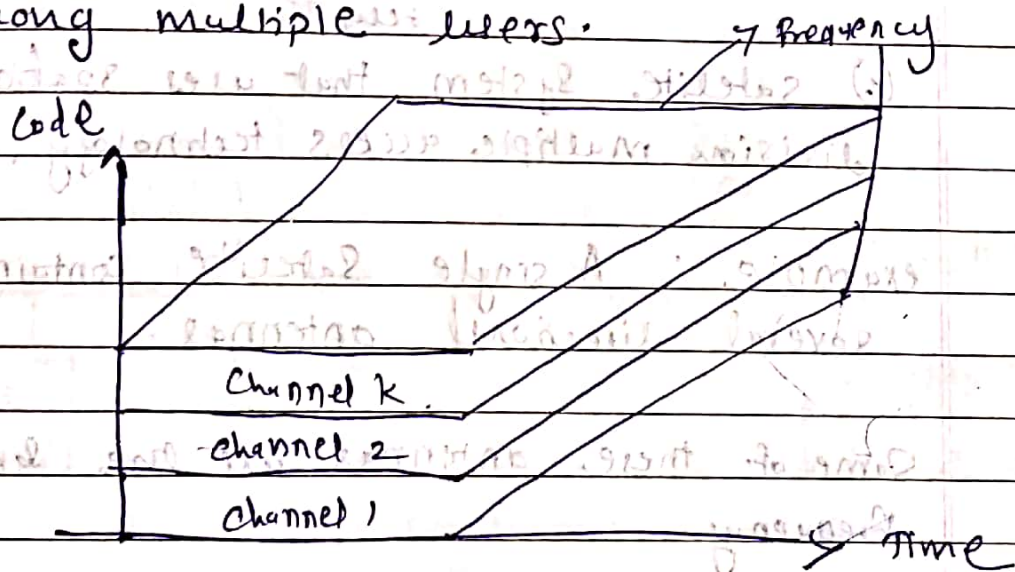


② Code division multiple access (CDMA)

① CDMA is a channel access method used by various radio communication technologies.

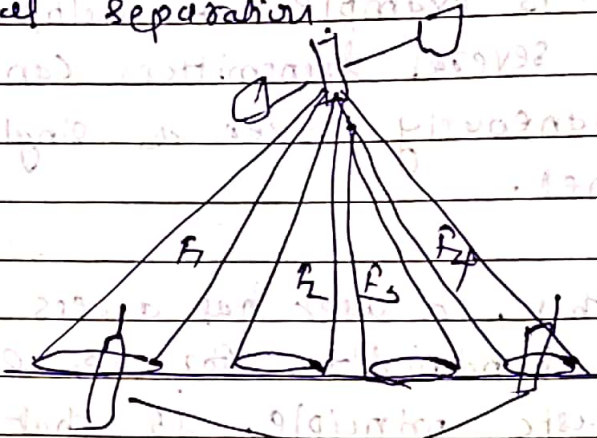
② CDMA is example of multiple access, where several transmitters can send inform simultaneously over a single communication channel.

③ In this, a user has access to the whole bandwidth for the entire duration. The basic principle is that different CDMA codes are used to distinguish among multiple users.



④ Space-division multiple access (SDMA)

① If there are no separation in frequency or time or code domain, an alternative way of separating different users is spatial separation.



Frequency reuse

② Satellite system that uses spatial division multiple access technology.

example : A single satellite contains several directional antennas.

Some of these antennas use the same frequency.

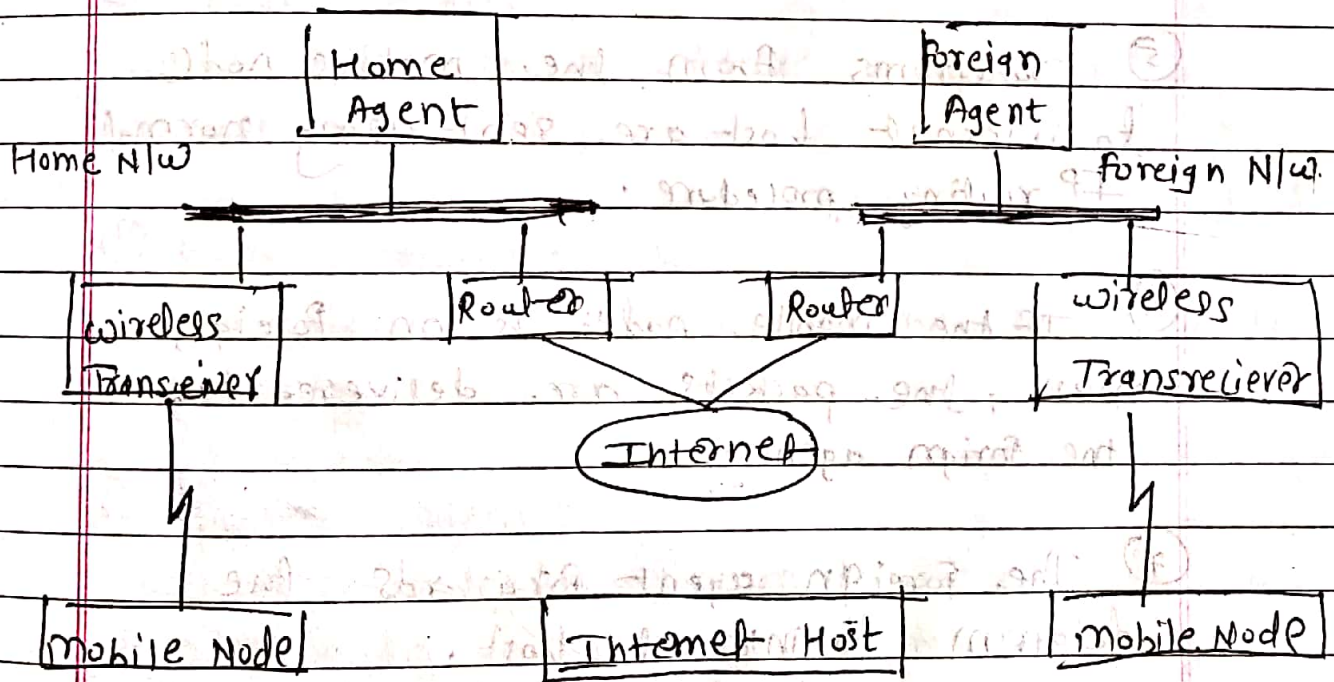
This allows a single satellite to simultaneously communicate to two different satellite receivers that operate on the same frequency.

- It is used for sharing of space in wireless n/w.
- Example of SDMA in wireless n/w is allocation of optimal base station to mobile phone user.

Mobile IP

This is an IETF (Internet Engineering Task Force) standard to communication protocol designed to allow mobile devices (such as laptop, PDA, mobile phone) users to move from one n/w to another n/w while maintaining their permanent IP (Internet Protocol) address.

Mobile IP Topology



OR

mobile IP is communication protocol that allows the users to move from one n/w to another with the same IP address.

- ① Internet host sends a datagram to mobile node using the mobile node's home address
- ② If the mobile node is on its home n/w, the datagram is delivered through the normal IP process to mobile node. & otherwise, the home agent picks up the datagram.
- ③ If the mobile node is on foreign network, the home agent forwards the datagram to the foreign agent.
- ④ The foreign agent delivers the datagram to the mobile node.
- ⑤ Datagrams from the mobile node to internet host are sent using normal IP routing procedure.
- ⑥ If the mobile node is on foreign n/w, the packets are delivered to the foreign agent.
- ⑦ The foreign agent forwards the datagram to internet host.

Components of Mobile IP.

① mobile Node (MN)

It is the hand-held communication device that the user carries eg cell phone.

② Home N/w

It is a n/w to which the mobile node originally belongs to as per its assigned IP address (home address).

③ Home Agent

It is router in home N/w to which mobile node was originally connected.

④ Home Address

It is permanent IP address assigned to the mobile node.

⑤ foreign network.

It is current n/w to which the mobile node is visiting.

⑥ foreign Agent -

It is a router in Foreign n/w to which mobile node is currently connected.

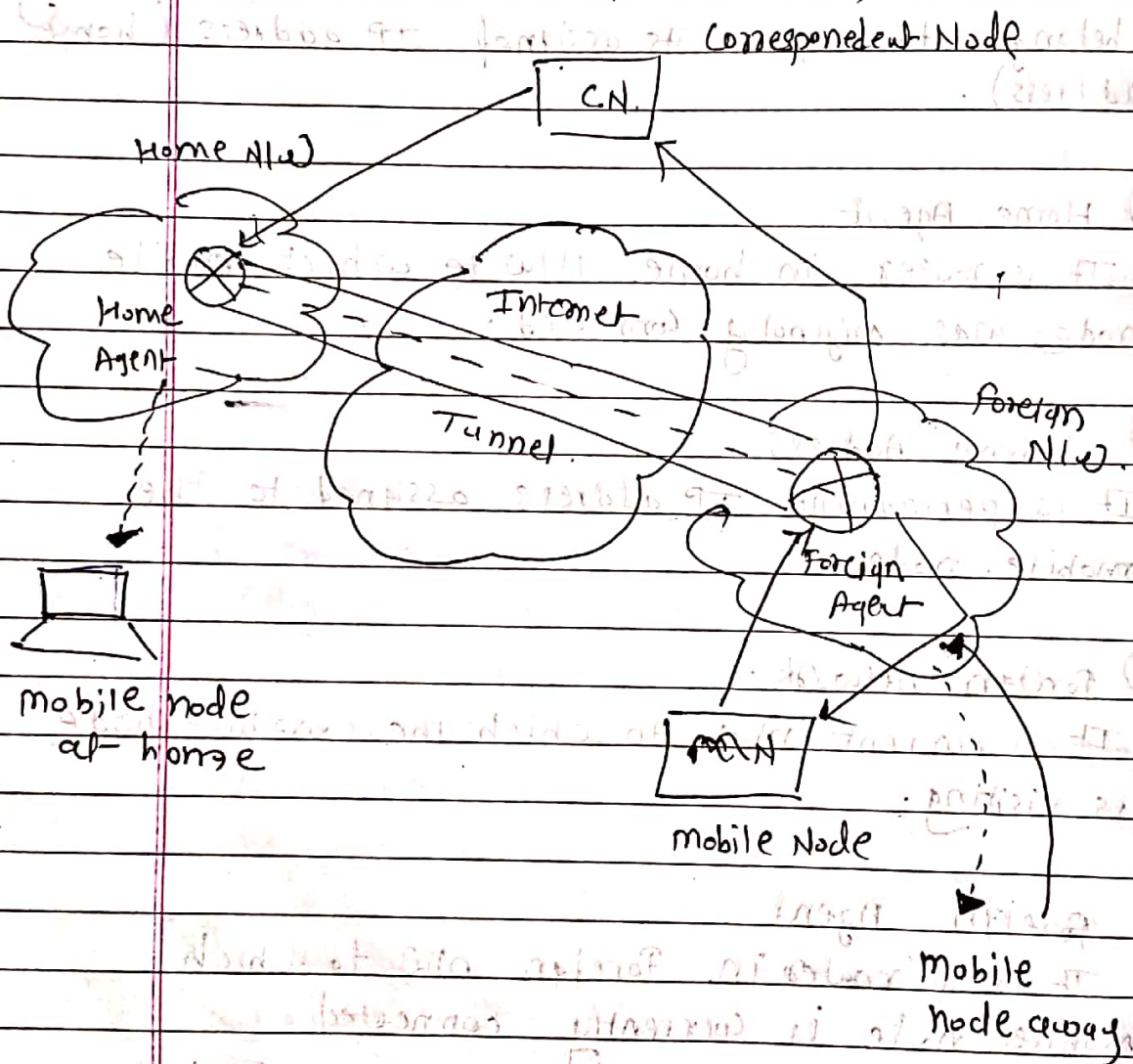
The packets from home agent are sent to Foreign agent which delivers it to the mobile node.

Correspondent Node (CN)

It is a device on internet communicating to the mobile node.

Care of Address (CoA)

It is temporary address used by a mobile node while it is moving away from its home n/w.



- ① Correspondent node sends the data to mobile node.
- ② Data Packets contains correspondent node address (source) and home address (destination).
- ③ Packets reached to the home agent.
- ④ mobile node is not in the home n/w, it has moved into the foreign n/w.
- ⑤ HA sends the care of address to home agent to which all packets should be sent.
- ⑥ Now tunnel will be established betⁿ home agent and foreign agent by process of tunneling.
- ⑦ Tunneling established a virtual pipe for the packets available betⁿ tunnel entry and endpoint.
- ⑧ It is the process of sending a packet via a tunnel and it is achieved by a mechanism is called encapsulation.
- ⑨ Home agent encapsulates the data packets into new packets in which the source address is home address and destination is care-of-address.
- ⑩ sends it through the tunnel to the foreign agent.
- ⑪ FA, on other side of tunnel received the data packets, decapsulated them and sends them to mobile node.
- ⑫ mobile node in response to data packets received, sends a reply in response to foreign agent.
- ⑬ foreign agent directly sends the reply to correspondent node.

Process of mobile IP

① Agent Discovery

During the agent discovery phase the HA and FA advertise their service on the network.

Mobile IP defines two methods:

① Agent advertisement

— Here HA and FA advertise their presence periodically using special agent advertisement message.

— These message advertisement can be seen as a broadcast into subnet.

✓ For this advertisement internet control message protocol (ICMP) are used with some mobility extensions.

② Agent solicitation

— If no agent advertisements are present or interval time is too high and an MN has not received a CoA, the mobile node must not send agent solicitations.

Registration

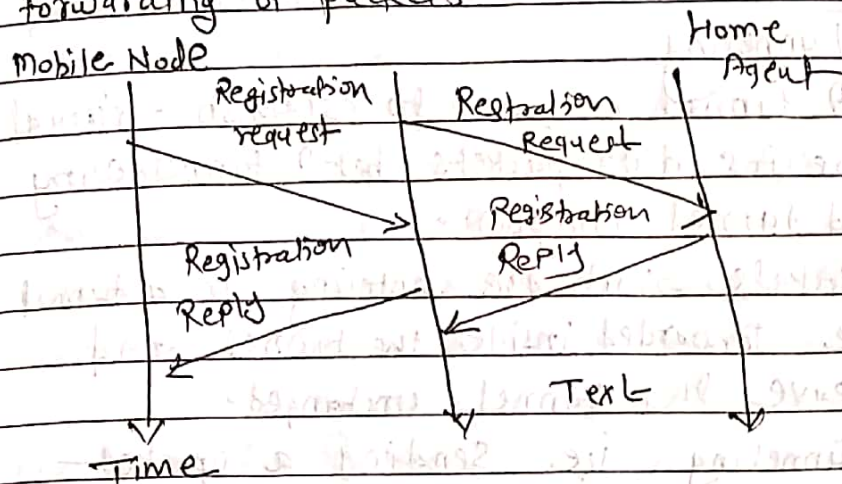
— ① In this phase, where MN registers its current location with FA and HA. ② If the connected location is identified as foreign location, then the MN looks for FA and registers itself with foreign location and FA, in turn, the HA ^{notifies} and creates a tunnel between itself and the FA. ③ During this phase, the MN sends registration request message to the FA which forwards the message to HA. ④ The HA sends back reply after updating its registration table with home address and care-of address mapping.

⑤ Thus successful mobile IP registration set up the routing mechanism for transporting packet from the mobile node.

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② Registration

The main purpose of registration is to inform the home agent of current location for correct forwarding of packets.



- If the CoA is at the FA, the MN sends its registration request containing the CoA to the FA which is forwarding the request to the HA.
- The HA now sets up a mobility binding containing the mobile node's home IP address and the current CoA.
- The mobility binding contains the lifetime of registration.
- Registration expires automatically after the lifetime and is deleted, so the mobile node should register before expiration.
- After setting up the mobility binding, the HA sends a reply message back to the FA which forwards it to the MN.

IF the CoA is co-located

The MN may send the request directly to HA.

③ Tunneling

① A tunnel is used to establish a virtual pipe for data packets betⁿ tunnel entry and tunnel endpoints.

② Packets which are entering in a tunnel are forwarded inside the tunnel and leave the tunnel unchanged.

③ Tunneling i.e. sending a packet through a tunnel is achieved with the help of encapsulation.

④ Tunneling is also known as "Port Forwarding" is the transmission and data intended for use only within private, usually corporate n/w through public n/w.

Routing (DSDV and DSR)

Routing is the process of finding the best path for the traffic in a network or across multiple networks.

Routing in a mobile ad-hoc network depends on many factors such as.

Modeling of topology, selection of routers, initiation of a route request.

Need for Routing.

- Route computation should not add many more nodes.
- If any host demands for the route, they must have quick access.
- In routing there must have a backup route when the.
- Maintenance of global state should not involve in route computation.

Routing Classification

① Proactive Protocol

- Proactive protocol attempt to evaluate continuously the routes within the network.
- It means proactive protocol continuously maintain routing information.
- So that when packet needs to be forwarded, the path is known already.

and can be immediately used.

Disadvantages :

① Requires more amounts of data for maintaining routing information.

② low reaction on restructuring n/w and failures of individual nodes

③ Reactive protocols

- Reactive protocols do not maintain routes but invoke a route determination procedure only on demand.

- or we can say reactive protocols build the routes only on demand.

- Thus when a route is required, some sort of global search procedure is initiated.

Advantages :

① No large overhead for global routing table

② Reaction is quick for n/w restructure and node failure.

③ Excessive Flooding can lead to n/w clogging.

④ Latency time is high in route finding

3) Hybrid Protocols

- Hybrid protocols attempt to take advantage of best of reactive and proactive schemes.
- The basic idea behind such protocols is to initiate route discovery on demand but at a limited search cost.

Routing Protocol

(1) Destination sequenced distance vector routing (DSDV).

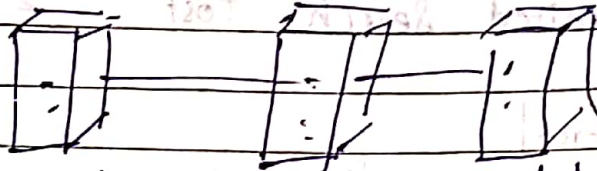
- It is table driven routing protocol for MANET based on Bellman ford algorithm.
- The main contribution of algorithm was that the algorithm works correctly, even in the presence of the loops in the routing table.
- Each mobile node maintains a routing table with route to every possible destination in the network.
- Each entry in the table contains a sequence number assigned by the destination node.
- The sequence number allow the node to distinguish stale routes from new and help avoid formation of routing loops.

A new route broadcast contains:

- ① The destination address.
- ② The number of hops required to reach the destination.
- ③ The sequence number of information received about the destination and new sequence number unique to broadcast.

- If there multiple routes are available for the same destination, the route with the most recent sequence number is used.

- If two updates have the same sequence number, the route with smaller metric is used to optimize the routing.



example. the routing table of Node A

From the above it is:

Destination	Next Hop	No. of hops	Sequence No.	Insert time
A	A	0	A46	001000
B	B	1	B38	001200
C	B	2	C28	001500

Advantage

- ① It is suitable for creating ad hoc network with small no. of nodes.

Disadvantage

- ① It's a regular update of its routing tables, which uses more battery power and small amount of bandwidth.

- ② This algo is not suitable for highly dynamic network.

② Dynamic Source Routing.

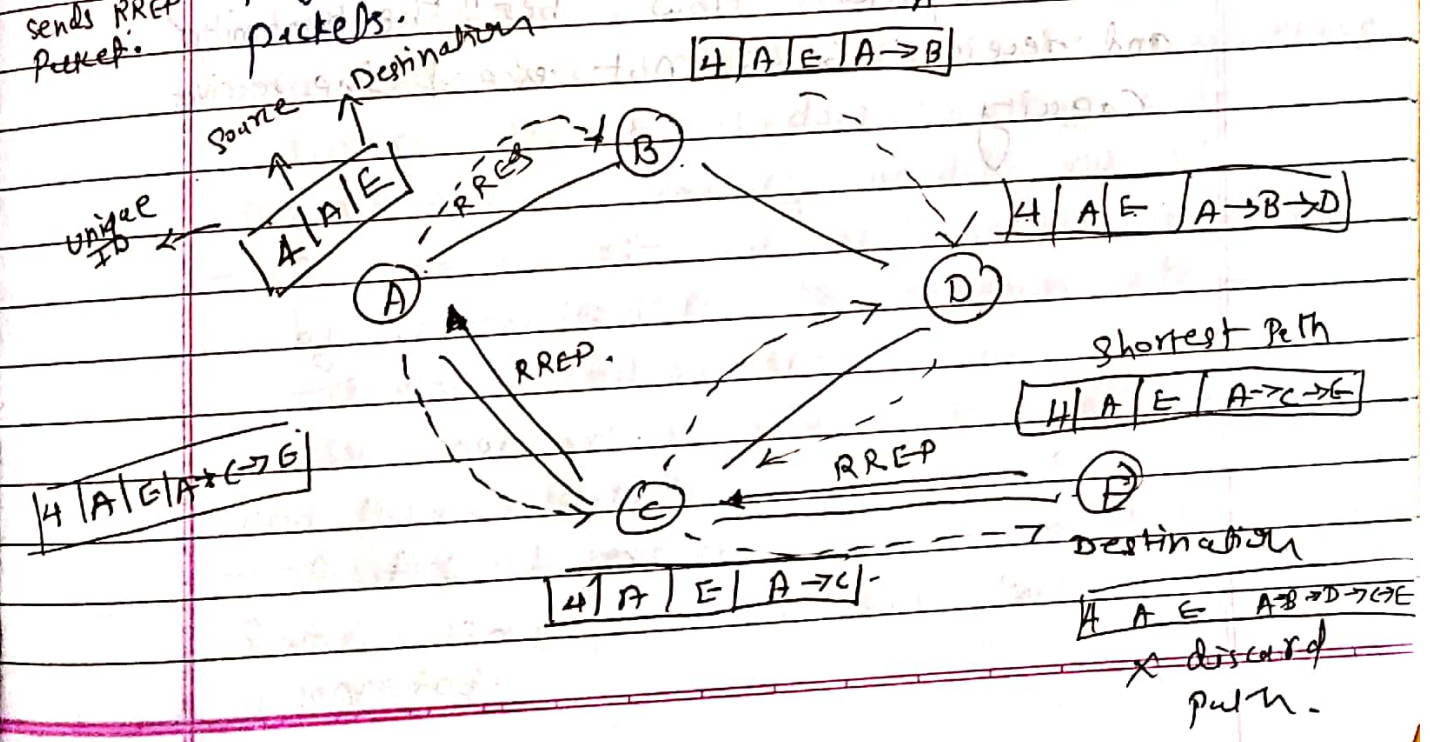
- DSR is an on-demand routing protocol which is based on source routing.
 - These algorithm maintain routes betn all the nodes even though there is no data exchange currently at all.
 - It may flood the net causing unnecessary traffic and also prevents the nodes from saving the battery power.
- Phases of DSR

① Route discovery

when there is no known route currently available, a node only discovers the route to destination when it wants to send to the destination. Two different packets RREQ. (Route Req). Basic ID's are source node ID and destination node ID.

② Route maintenance

RR (Route Reply) A node has to ensure that the route is uprigh when it is continuously sending packets. A → B route request

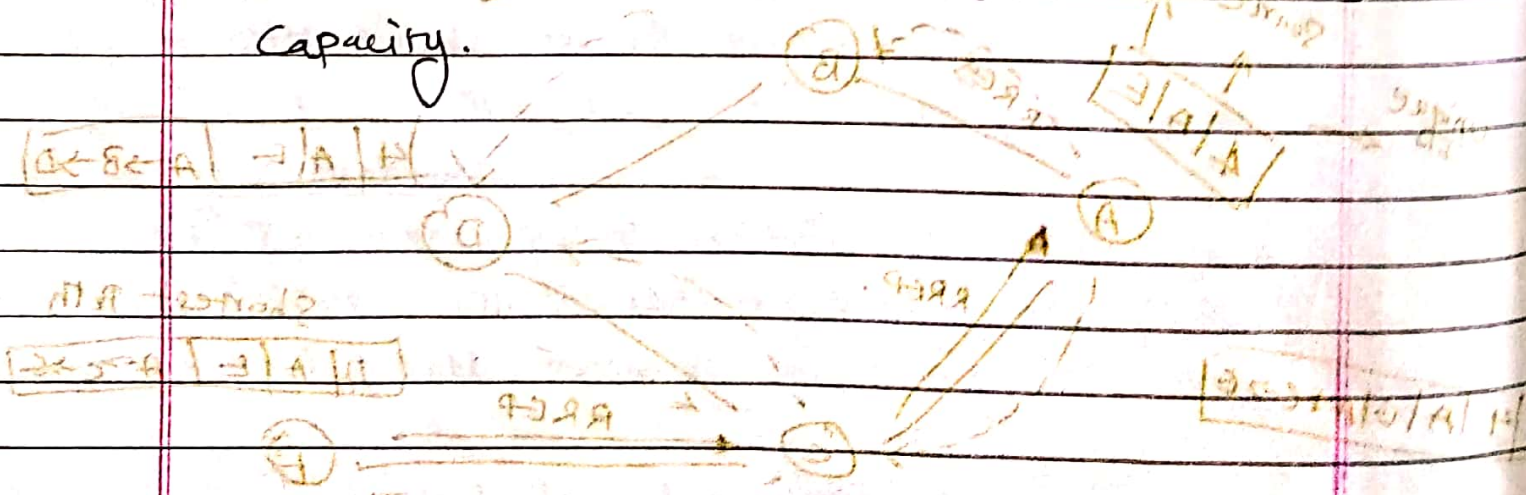


Mobile TCP

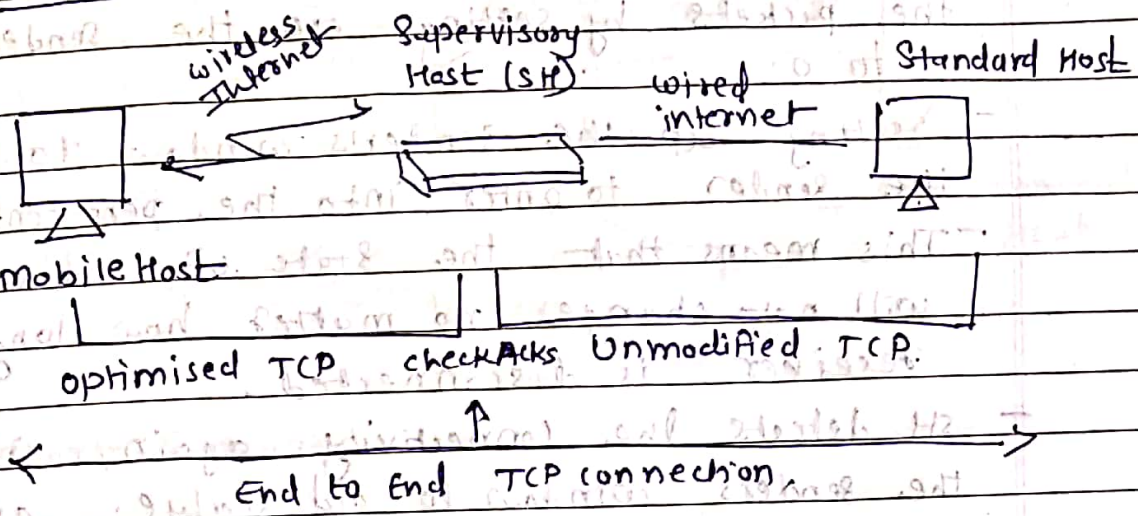
- Tradition TCP (Transmission Control Protocol) is one of the core protocols of internet protocol.
- TCP is reliable, guarantees in order delivery of data and incorporates congestion control and flow control mechanisms.
- TCP supports many of the internet's most popular application protocol and resulting applⁿ, FTP and secure shell.
- In internet protocol, TCP is the intermediate layer between the internet layer & application layer.

Responsibilities

- ① Provide reliable in-order transport of data. To not allow losses of data.
- ② Control congestions in the net: to not allow degradation of net performance.
- ③ Control packet flow betⁿ the transmitter and receiver. To not exceed the receiver capacity.



mobile TCP



- mTCP partitions the TCP connection into two parts (1) unmodified TCP, (2) optimized TCP.
- unmodified TCP is used on the standard host supervisory host connection.
- Optimized TCP is used on standard host-mobile host connection.
- SH controls the exchange of data between both parts.
- m-TCP has low bit-error rate on wireless links. Hence, it does not carry out retransmission or caching of data via SH.
- If packets lost, it has to be retransmitted by original sender. This maintains the TCP end to end semantics.
- SH monitors all packets sent to the MH and Ack's returned from the MH.
- If Ack not returned received by SH for some time, it assumes that MH is disconnected.

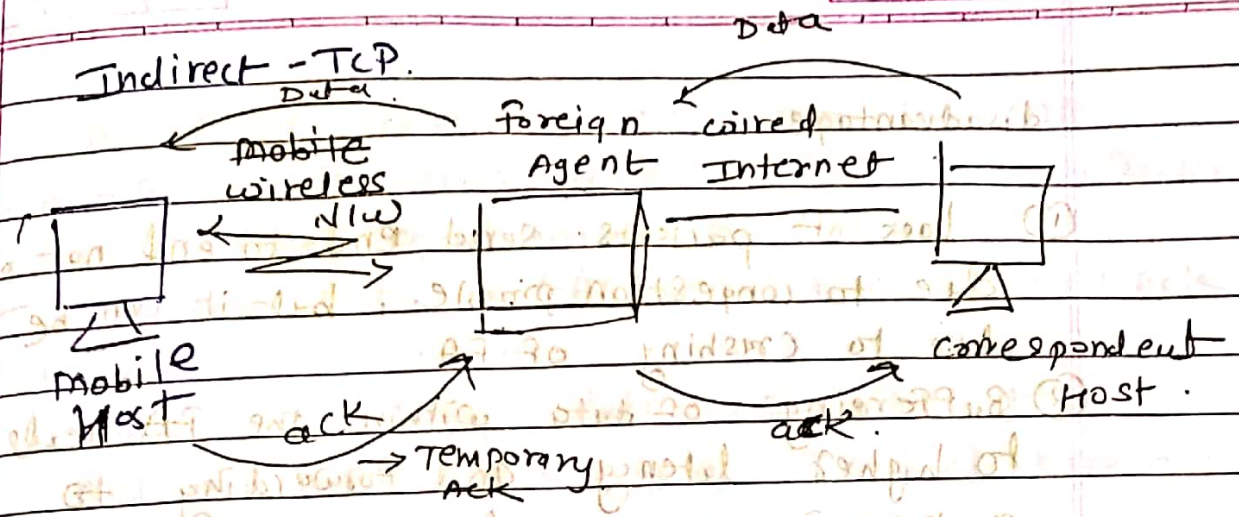
- it then prevents sender from transmitting the packets by setting up the senders window to 0.
- setting up the sender's window to 0 forces the sender to enter into the persistent mode.
- This means that the state of the sender will not change no matter how long receiver is disconnected.
- It detects the connectivity again, it reopens the senders window to old value. and now sender can continue sending with full speed. This doesn't require changes in the sender's TCP.
- The wireless side uses adapted TCP that can recover from packet loss much faster.

Advantages

- ① Does not require the change in the sender's TCP.
- ② No need to send old buffer data to new SH.
- ③ IF MH disconnected it doesn't waste time in useless transmissions and shrinks the window size to 0.

Disadvantages

- ① Modifications are required for MH protocol.
- ② When a number of nodes move to new SH, the bandwidth need to be managed.



Short Summary.

DET is optimised TCP

- ② CH sends data to FA, FA will send Ack.
- ③ CH recognised the data with MH.
- ④ FA sends data to MH.
- ⑤ MH sends ack to FA. (Temporary Ack)
means not sends to CH.
- ⑥ IF MH not send ACK then means data is not sent to MH.
- ⑦ so FA will retransmits the data to MH
is called as indirect TCP.

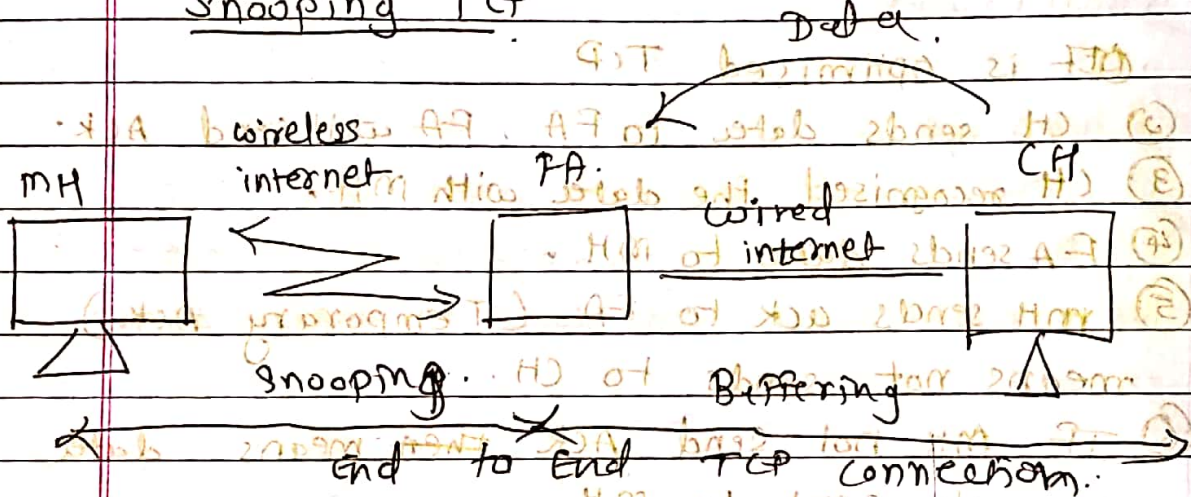
Advantages

- ① Does not require changes to TCP at the host in the fixed net.
- ② Errors from the wireless link are corrected at TCP proxy.
- ③ New protocol affects only a limited part of internet.
- ④ optimization possible over a wireless link.

disadvantage

- ① Loss of packets send end-to end not only due to congestion problem, but it can be due to crashing of FA.
- ② Buffering of data within the FA leads to higher latency and forwarding to a new foreign agent.

Snooping TCP



- ① Snooping provides the end-to end TCP connection.
- ② So here we added snooping TCP approach
- ③ Snooping watch the every packets in both the directions.
- ④ Snoop layer look at every packet in connection.
- ⑤ Buffering stores the data.
- ⑥ CH sends data to FA. FA not send ack to CH.
- ⑦ FA stores (buffer) the data.

- (8) FA sends data to MH one by one.
example - (CH sends data to FA like D_1, D_2, D_3 but FA will send only D_1 to MH)
- (9) MH send ACK to FA and FA will send ACK to CH.
- (10) Now FA sends data to MN but MN not send ACK then FA retransmit the data.
- (11) Then MN will send ACK to CH data will get.

Advantage :

- (1) End to end semantics are preserved.
- (2) CH need not to be changed; most of the enhancements are done in foreign agent.

DSDV (Continue)

In this each node keeps records of route infoⁿ in the form of routing table

Table consist of

- Destination ID

- next node

- distance (No of hops)

- sequence no

Route broadcast message

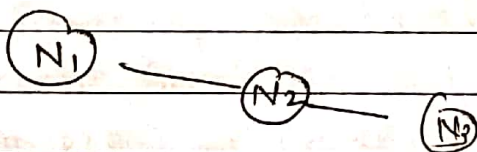
- Destⁿ node

- next hop

- Recent seq no

- Distance

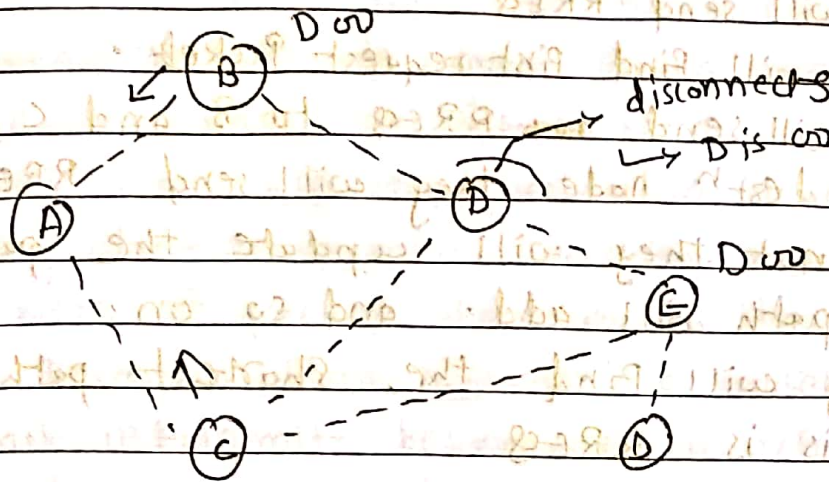
Each node exchanges its updated routing table with each other.



Routing table of N₁

Dest	next node	distance	Seq.
N ₂	N ₂	1	14
N ₃	N ₂	2	18

Table maintenance in DSDV.



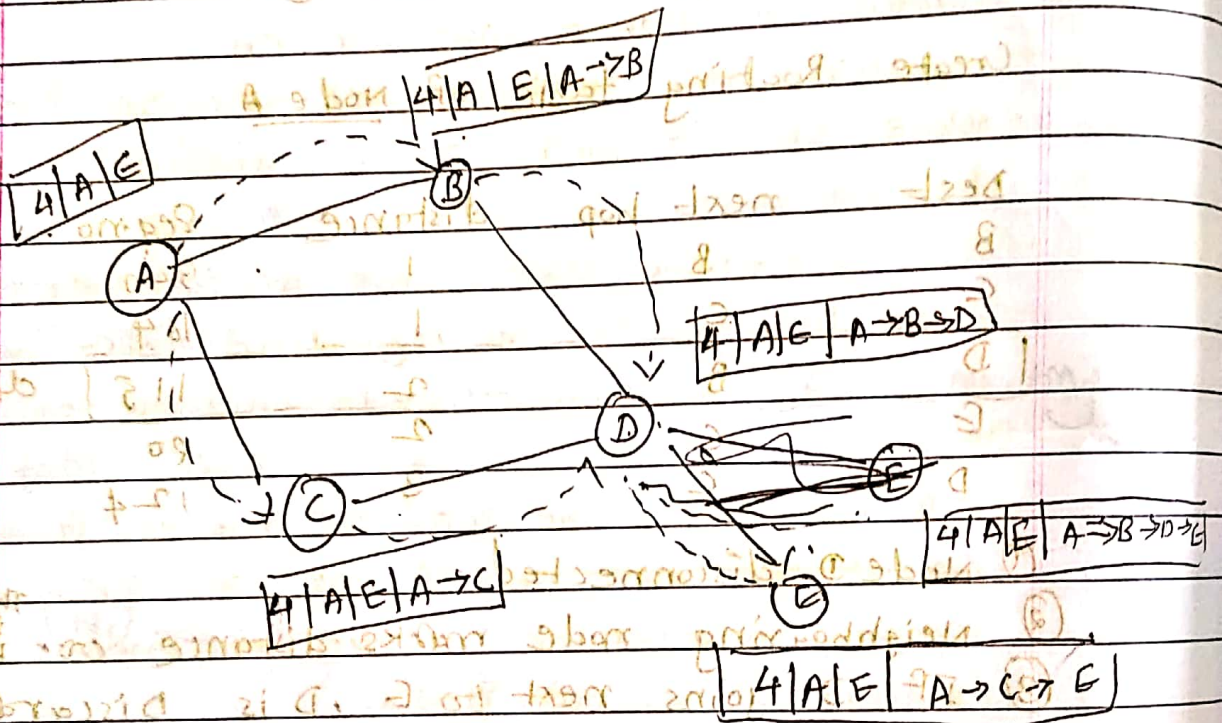
Create Routing table of Node A

Dest	next hop	distance	Seq no
B	B	1	340
C	C	1	164
D	B	2	115
E	C	2	20
D	C	3	124

They will broadcast to its neighbor

- ① Node D disconnected
- ② Neighbouring node marks distance ∞
- ③ If D joins next to E, D is Discarded from maintainable
- ④ The new entry will be added in RT's and the sequence number should be greater than first one. (115 to 124)

- ① A is source, E is destⁿ
- ② A will send RREQ.
- ③ They will find first request packet.
- ④ A will send RREQ to B and C, IF they are destⁿ node, they will send RREP,
- ⑤ IF not they will update the packets so path will add. and so on.
- ⑥ They will find the shortest path. (Optimum path)
- ⑦ This is RREQ



Fast Retransmit / Fast Recovery

- Whenever a MH moves to a new FA, it can cause packet loss at CN.
- * - with the help of mechanism like fast recovery / fast retransmit, a host can detect the reason of packet loss other than congestion after reception of duplicate acknowledgments.
- Fast retransmit behaviour is artificially invoked on mobile host and CN.
- when the MH registers at new FA using mobile IP, it starts sending duplicated ack to CN.
- It is proposed that three ack needs to send.
- This forces the corresponding host to go into fast retransmit mode and not to slow start.
- * In this case, the sender moves to new FA, it can cause packet loss or slow start mechanism assuming congestion even though congestion is not present.
- Therefore CN starts sending with same rate as it was doing before MH moved on new FA.
- This approach puts MH in the fast retransmit mode to avoid slow start at MH after moving to new FA.

Selective Retransmission

- when some of the packet is lost in TCP, the sender has to retransmit everything starting from lost packet.
- such retransmission of all chunks of data wastes bandwidth.
- This issue can be resolved by using selective retransmission.
- In this scheme, TCP can indirectly request a selective retransmission of packets.
- The receiver can ack single packets and not the train of packets.
- Hence sender can precisely determine which packet is lost and can retransmit only the lost one.